Product Master Data

Strategic Architecture & Governance Reference

Version 2.0 | Module: All | Odoo 19

Kebony — Custom Platform Architecture

Why Product Matters

The product record is a load-bearing data structure — 13 domains depend on it

#	DOMAIN	WHAT PRODUCT DRIVES
1	**Sales**	Quotation, pricing, packaging UoM
2	**Pricing**	Pricelist rules, price per lm / m^2 / m^3
3	**Inventory**	FIFO valuation, lot tracking, quant metrics
4	**Manufacturing**	BoM generation, dryer load planning, chemical mix
5	**Capacity Planning**	Dryer / autoclave scheduling, bottleneck
6	**Costing & Margin**	Standard cost, COGS split, margin per m^3
7	**Accounting**	GL routing, analytic distribution, accruals

#	DOMAIN	WHAT PRODUCT DRIVES
8	**Logistics**	Weight, pallet dims, transport planning
9	**Packaging**	Pack / Board UoMs, picking, shipping labels
10	**Compliance**	HS codes, intrastat, customs declarations
11	**Website**	Catalogue, filtering, search, display
12	**Purchasing**	Vendor bills, consignment, raw material link
13	**Samples**	Sample workflows, services available

> A single field error propagates across inventory valuation, costing, invoicing, logistics, and compliance.

Architecture Principles

Four structural rules that shape the entire product model

1. No Dimensional Variants

Odoo 19 merged packaging and UoM. A product sold as "Pack of 162 × 3.2 lm" needs packaging-level UoMs that depend on length. Each dimensional product = its own `product.template`.

2. Metric / Imperial Separation

Metric and imperial are **separate templates** linked by a shared Master Item Code. Metric is the engineering source of truth; imperial derives from it.

CONCEPT	METRIC	IMPERIAL
Name	`1127-M-3.2`	`1127-I-10.5`
UoM	m (meter)	LF (linear foot)

3. Treatment ≠ Variant

Treatment (fire retardant, colour, brushing) is tracked at **lot level only**. Product identity remains purely dimensional.

Without this rule: ~70 cross-sections × ~20 lengths × 4 treatments × 2 systems = **~11,200 SKUs**. Current catalogue: **~490**.

4. Three-Level Hierarchy

```
Master Item (e.g. 1127)
  └── Master Variant (e.g. 1127-3.2)
        ├── Metric Product  (1127-M-3.2)
        └── Imperial Product (1127-I-10.5)
```

Product Population & Hierarchy Levels

~490 products across two wood species, two measurement systems, plus accessories & services

PRODUCT POPULATION

SEGMENT	METRIC	IMPERIAL	TOTAL
Character (Scots Pine)	30	~30	~60
Clear (Radiata)	117	117	~234
Master Items	~70	—	~70
Master Variants	~80	—	~80
Accessories	—	—	~27
Services / Other	—	—	~20
Estimated Total			**~490**

HIERARCHY LEVELS

Master Item NON-SELLABLE

`x_studio_is_master_item = True`

Cross-section grouping (e.g. "Kebony Character Decking 28×120")

Master Variant NON-SELLABLE

`x_studio_is_master_variant = True`

Dimension grouping (e.g. "1127-3.2" = 28×120 at 3.2m)

M-I Master LINKING RECORD

`x_studio_product_family_1 = True`

Links metric/imperial twins (e.g. `1127-M`)

Sellable Product STOCKABLE

Actual `product.template` (e.g. `1127-M-3.2`)

Measurement Doctrine

Three dimensions of quantity & the critical volume distinction

THREE DIMENSIONS OF QUANTITY

Linear OPERATIONAL

meters (m), lm, LF — normalised to meters internally

Board Count INFORMATIONAL

Integer boards — ROUND_UP at threshold 0.95

Volume ECONOMIC TRUTH

m³ — all KPIs, pricing, margin analysis in m³

CONVERSION CONSTANTS

```
FT_PER_M    = 3.280839895
M_PER_FT    = 1 / FT_PER_M
FT3_PER_M3  = 35.3146667
```

CRITICAL — TWO VOLUME FIELDS

FIELD	TECHNICAL NAME	UNIT	USAGE
Normalised	`x_studio_volume_m3`	Always m³/lm	CANONICAL All economic calcs
Localised	`volume` (Odoo)	m³/lm or ft³/lf	TRANSPORT ONLY Never for metrics

Formula: Total volume = linear_meters × x_studio_volume_m3
Never use `product.volume` for economic calculations.

KEY FORMULAS

DERIVED FIELD	FORMULA
Volume (m³/lm)	thickness(m) × width(m)
lm per m³	1 / `x_studio_volume_m3`
lm per m²	1 / width(m)
Weight per lm	density × volume(m³/lm)
Boards	ROUND(linear_m / length_m), threshold 0.95
Total volume	linear_m × `x_studio_volume_m3`

Field Registry — Core Identity

18 fields that define what a product is — Odoo standard + Studio-defined

#	LABEL	TECHNICAL FIELD	TYPE	NOTES
1	Name	`name`	Char	`{Item}-{System}-{Length}` e.g. `1127-M-3.2`
2	Description	`x_studio_description`	Char	Full commercial name
3	Sales Description	`description_sale`	Text	Shown on quotations
5	Master Item Code	`x_studio_master_item_code`	Char	Cross-section grouping: `1127`
6	Is Master Item?	`x_studio_is_master_item`	Bool	True for grouping records
7	Master Variant Code	`x_studio_master_variant_code_1`	Char	Dimension grouping: `1127-3.2`
9	M-I Master Code	`x_studio_product_family`	Char	Links metric/imperial: `1127-M`
11	Measurement System	`x_studio_measurement_system`	Selection	`Metric` · `Imperial`
12	Unit of Measure	`uom_id`	Many2one	`m` (metric) or `LF` (imperial)
14	Product Category	`categ_id`	Many2one	GL routing (accounting)
15	Product Type	`type`	Selection	`Goods` · `Service` · `Combo`
16	Lot Tracking	`tracking`	Selection	`By lots` for all wood
17	Valuation by Lot	`lot_valuated`	Bool	True — FIFO per lot

Naming convention: Fields 1–4 are Odoo standard | Fields 5–11 are `x_studio_*` (Studio-defined) | Fields 12–18 are Odoo standard

Field Registry — Classification & Geometry

Classification attributes + metric engineering base dimensions

CLASSIFICATION FIELDS

#	LABEL	TECHNICAL FIELD	NOTES
19	Wood Species	`x_studio_word_species`	`Scots Pine` · `Radiata`
20	Production Family	`x_studio_production_chain`	`Character` · `Clear`
21	Application	`x_studio_many2one_field_3oj…`	7 values (M2O)
22	Shape	`x_studio_shape`	27 values (M2O)
24	Product Lifecycle	`x_studio_product_lifecycle`	6 stages
25	Product Classification	`x_studio_product_classification`	`wood` · `accessory` (M2M)

Note: `x_studio_word_species` — the typo ("word" not "wood") is permanent. Studio fields cannot be renamed.

METRIC GEOMETRY (ENGINEERING BASE)

#	LABEL	TECHNICAL FIELD	UNIT
27	Length	`x_studio_length_m`	m
28	Thickness	`x_studio_thickness_mm`	mm
29	Width	`x_studio_width_mm_1`	mm
30	Volume (normalised)	`x_studio_volume_m3`	m^3/lm
31	Weight	`x_studio_weight_kg`	kg
32	Density	`x_studio_density_kgm3_2`	kg/m^3
33	Square Factor	`x_studio_square_factor_1`	lm/m^2
35	lm per m^3	`x_studio_number_of_lmm3_1`	lm/m^3
36	lm per m^2	`x_studio_number_of_lmm2_1`	lm/m^2
37	Weight per lm	`x_studio_net_weight_per_lm_kg_1`	kg/lm

All derived fields must be reproducible from the **four base dimensions**: length, thickness, width, density.

Field Registry — Imperial Geometry & Packaging

Imperial values derive from metric + packaging intelligence fields

IMPERIAL GEOMETRY

#	LABEL	TECHNICAL FIELD	UNIT
39	Length (LF)	`x_studio_length_1`	feet
40	Thickness	`x_studio_thickness`	inches
41	Width	`x_studio_width`	inches
42	Volume (localised)	`volume`	ft^3/lf
43	Weight	`weight`	lb
44	Density	`x_studio_density_lbft3`	lb/ft^3
45	Square Factor	`x_studio_square_factor`	lf/ft^2
47	LF per ft^3	`x_studio_number_of_lmm3`	lf/ft^3
49	Weight per LF	`x_studio_net_weight_per_lm_kg`	lb/lf

PACKAGING INTELLIGENCE

#	LABEL	TECHNICAL FIELD	TYPE
53	Boards per Pack	`x_studio_boards_per_pack`	Integer
54	Pcs per Layer	`x_studio_pcslayer`	Integer
55	Number of Layers	`x_studio_number_of_layers_1`	Integer
56	Pallet Height (cm)	`x_studio_pallet_height_cm_1`	Float
57	Pallet Width (cm)	`x_studio_pallet_width_cm_1`	Float
59	End-Matched	`x_studio_end_matched`	Selection
61	Packagings (UoMs)	`uom_ids`	M2M

UoM Packaging Example

```
Product: 1127-M-3.2
UoMs:    m, Board of 3.2 lm, Pack of 162 × 3.2 lm

End-Matched values: End-matched · End-cut · N.A.
Orientation:        Horizontal · Vertical · Both · N.A.
```

Supply Chain Rules

Four replenishment strategies — per-product assignment based on market maturity

Make to Order

Buffer: None
Trigger: Min. process batch reached
Lead time: Procurement + production + transport

New / low-volume products

ATO — White Wood

Buffer: White wood (untreated)
Trigger: Min. white wood stock
Lead time: Production + transport

Start from autoclave stage

ATO — Brown Wood

Buffer: Brown wood (treated RS)
Trigger: Min. brown wood stock
Lead time: Machining + transport

Radiata — machining only

Make to Stock

Buffer: Finished goods
Trigger: Min. FG stock breach
Lead time: Transport only

High-volume / mature

```
MTO:     [Procurement] → [Stacking] → [Autoclave] → [Dryer] → [Machining] → [Transport]   ← full lead time

ATO-WW:               [White Wood Buffer] → [Autoclave] → [Dryer] → [Machining] → [Transport]

ATO-BW:                                     [Brown Wood Buffer] → [Machining] → [Transport]

MTS:                                                              [FG Buffer] → [Transport]
```

Products can transition: MTO (introduction) → ATO → MTS (mature) as demand stabilises. Per-product assignment awaiting COO input.

Product Categories & Trade Compliance

Accounting routing via category + HS codes for customs & intrastat

PRODUCT CATEGORIES (GL ROUTING)

CATEGORY	ACCOUNTING IMPACT
Finished Product - Wood	FIFO valuation, standard COGS
Semi-Finished Product - Wood	WIP valuation
Raw Material - White Wood	Purchase → WIP conversion
Raw Material - Chemical	Consumed in BoM
Finished Product - Accessories	Separate COGS category
Product in Consignment	Consignment valuation rules
Sample Special / Standard	Sample cost tracking
Services	No inventory impact
Landed Costs	Cost adjustment layer

HS CODE & TRADE CLASSIFICATION

PRODUCT TYPE	HS (NO)	HS (EU)	ORIGIN
Radiata Rough Sawn	44.07.1111	4407 11 90 90	BE / NO
White Radiata RS	44.07.1121	4407 11 90 90	NZ
Radiata Machined	44.09.1091	4409 10 18 00	BE / DK
Character ≤48×48	44.09.1091	4409 10 18 00	NO
Character Standard	44.09.1092	4410 10 18 00	NO
White Scots Pine	44.09.1091	4409 10 18 00	SE
White SYP	44.07.1121	4407 11 90 90	US

COMPLIANCE FIELDS

LABEL	TECHNICAL FIELD
HS Code	`hs_code`
Commodity Code	`intrastat_code_id`
Country of Origin	`intrastat_origin_country_id`

Planning Family Model

Bridge between product and manufacturing — `kebony.planning.family`

PRODUCT → FAMILY LINK

#	LABEL	RO
79	Planning Family	EDIT
80	Stacking Hours	READ
81	Autoclave Hours	READ
82	Dryer Hours	BOTTLENECK
83	Dryer h/m³	READ
84	Typical Load m³	READ
85	Min Process Qty	READ
86	Mix kg/m³	READ
87	Internal Scrap %	READ
88	B-Grade Scrap %	READ

> Only **#79** (Planning Family) is editable on the product. All others are related / readonly from the family record.

FAMILY INVENTORY

CODE	NAME	TYPE	DRYER	AC	H/M³	MIX
KSP1C	SP 1" Cladding	SP	55	3.5	1.34	230
KSP15T	SP 1.5" T&G	SP	55	3.5	—	—
KSP2PD	SP 2" Decking	SP	65	3.5	1.57	315
KSP2C	SP 2" Cladding	SP	65	3.5	—	—
KR1	Radiata 1"	R	54	4.5	1.18	676
KR1CL	Radiata 1" Cladding	R	54	4.5	—	—
KR2	Radiata 2"	R	62	4.5	1.36	600
KRRS1	Radiata RS 1"	R	—	—	—	—
KRRS2	Radiata RS 2"	R	—	—	—	—
KSPRUCE	Kebony Spruce	O	30	3.5	0.77	40

SP = Scots Pine, R = Radiata, O = Other. Dryer/AC in hours. Mix = kg/m³.

Metrics Computation Architecture

The mixin, recompute hierarchy, and wood-only rule

THE MIXIN: `KEBONY.METRICS.MIXIN`

Abstract model inherited by: `stock.quant`, `stock.lot`, `stock.move`, `sale.order.line`, `account.move.line`, `product.template`.

METHOD	PURPOSE
`_kebony_linear_qty_in_m()`	Normalise any quantity to meters
`_kebony_boards_from_linear()`	Board count (ROUND_UP 0.95)
`_kebony_volume_m3()`	Volume via `x_studio_volume_m3`
`_kebony_is_wood_product()`	Check classification for "wood"

Wood-Only Rule: Physical metrics (linear m, boards, volume m³) are **only computed for wood products**. Accessories and services return 0.

RECOMPUTE HIERARCHY

```
Level 1: stock.quant      ← source of truth
           ├→ stock.lot          (aggregate from internal quants)
           └→ product.template   (aggregate from internal quants)

Level 2: stock.move            (from move lines or demand qty)
         sale.order.line       (from SO line qty + UoM)
         account.move.line     (from invoice line qty + UoM)

Level 3: sale.order            (sum of order lines)
         account.move          (sum of invoice lines)
```

COMPUTED STOCK METRIC FIELDS

FIELD	MODEL	STORED
`x_studio_number_of_boards`	product.template	Yes
`x_kebony_volume_m3_on_hand`	product.template	Yes
`x_kebony_available_packs`	product.template	Yes
`kebony_boards_available`	product.template	No
`kebony_cubic_meters_available`	product.template	No

Entity Type & Company Architecture

Shared products, entity-gated features — `res.company.x_kebony_entity_type`

COMPANY REGISTRY

CODE	COMPANY	TYPE	FEATURES
`inc`	Kebony Inc. (US)	Distribution	BOL, accruals, margin, consignment
`holding`	Kebony Holding (BE)	Holding	Product master ownership
`bnv`	Kebony BNV (BE)	Manufacturing	Planning, BoM, manufacturing
`kas`	Kebony KAS (NO)	Legacy	Placeholder
`nas`	Kebony Norge AS (NO)	Legacy	Placeholder

Key Methods

```
_is_kebony_us()   → True for INC
_is_kebony_bnv()  → True for BNV
```

FEATURE GATING BY ENTITY

US-Only Features (INC)

- Bill of Lading (BOL) reports
- Accruals (royalties, mgmt fees, freight, duty)
- Margin analysis per m³
- Consignment / Biewer PO flow
- Imperial packaging & labels

Manufacturing Features (BNV)

- Planning families & process parameters
- Dryer-centric batch planning
- Bill of Materials generation
- Production scheduling

Global Features (All Entities)

- Product master data & metrics
- KebonyMetricsMixin calculations
- Inventory & lot tracking

Traceability & FIFO Integrity

Lot-level identity, treatment separation, and per-lot cost tracking

Lot-Level Identity

Each physical production batch receives a unique `stock.lot` record carrying:

- Treatment info (fire retardant, colour, brushing)
- Metrics: linear meters, boards, volume
- Pack reservation: link to reserved SO
- Legacy fields: `x_studio_boards`, `x_studio_linear_feet`

Product ≠ Treatment

Treatment is a **post-production attribute** applied during or after manufacturing.

It is **never** part of the product identity or SKU. This prevents combinatorial SKU explosion.

Product identity = dimensional specification only.

FIFO Integrity

Odoo 19's lot-valuated FIFO ensures per-lot cost tracking:

- `lot_valuated = True`
  → per-lot cost tracking
- `tracking = 'lot'`
  → mandatory lot assignment
- `categ_id`
  → GL account routing for valuation

WOOD SPECIES & PRODUCTION FAMILIES

SPECIES	LATIN NAME	ORIGIN	PRODUCTION FAMILY
Scots Pine	*Pinus sylvestris*	Sweden / Norway	Character
Radiata	*Pinus radiata*	New Zealand / Belgium	Clear

PRODUCT LIFECYCLE STAGES

INTRODUCTION | GROWTH | MATURE | DECLINE | ABANDON | DECOMMISSIONED

Lifecycle transitions are one-way. A product cannot move from Decommissioned back to Mature without executive approval.

Reference Values & Classifications

Model values for Application, Shape, Market Portfolios, and Services

APPLICATION (7 VALUES)

VALUE	TYPICAL PRODUCTS
Decking	Floor boards, boardwalk planks
Cladding	Wall profiles, façade boards
Construction	Structural beams, battens
Boardwalk	Heavy-duty outdoor planking
Roofing	Roof boards
Industrial	Non-standard / custom
N.A.	Raw materials, semi-finished

MARKET PORTFOLIOS

EXPORT · FRANCE · NORDIC · DACH · BENELUX · UNITED STATES

Products can belong to multiple markets (M2M).

SERVICES AVAILABLE

FIRE RETARDANT · COLOR FINISH · BRUSHING · POLISH · SHOU SUGI BAN · FINE SAWN

SHAPE (27 VALUES)

Batten · Beam · Bevel Halflap · Bevel T&G · Comb Faced · Dekora T&G · Halflap · Melsom Slat · N.A. · Nickel Gap · Railing · Rectangular · Rectangular Click-In · Rectangular w Side Slits · Rhombus · Rhombus Click-In · Rhombus Tiga · Roof Board · Rough Sawn · Shiplap · Shiplap Click-In · Siding · Skewed Slat · StepClip · StepClip Startboard · T&G · Triple Batten

SHAPE VALUE (27 PROFILE GEOMETRY CODES)

`1SL` · `1SLL` · `1SLR` · `2SL` · `3SL` · `45°` · `60°` · `60° 1SL` · `75°` · `75° 1SL` · `75° 1SLL` · `75° 1SLR` · `75° 2SL` · `75° 3SL` · `78°` · `78° 1SL` · `90°` · `90° 1SL` · `90° 1SLR` · `90° 2SL` · `C1` · `C2` · `C3` · `N.A.` · `R2` · `R3` · `R5`

FIXING METHOD (8 VALUES)

`[blank]` · Click-In · Nickel Gap · Side Slits · Dekora · Tiga · GRAD · CAMO

Governance Framework

Master Data Guardian, RACI, change control & critical rules

MASTER DATA GUARDIAN

A **Super Master Data Guardian** must be nominated — single point of accountability for product data integrity across the organisation.

- **Authority:** No product CRUD without Guardian validation
- **Scope:** All entities, all product types, all governed fields
- **Cadence:** Weekly review, monthly audit

CHANGE CONTROL PROCESS

```
1. REQUEST    → Fields, current → proposed, justification
2. REVIEW     → Cross-domain & financial impact
3. APPROVE    → Guardian sign-off (or escalate)
4. EXECUTE    → Applied in Odoo
5. VALIDATE   → Metrics, orders, planning intact?
6. DOCUMENT   → Change logged in register
```

RACI MATRIX

FIELD FAMILY	R	A	C	I
Identity	MDG	COO	Sales	All
Geometry — Metric	CTO	MDG	Prod.	Finance
Geometry — Imperial	CTO	MDG	US Sales	Finance
Packaging	Planning	MDG	Logistics	Sales
Classification	Prod. Mgmt	MDG	Eng.	Mktg
Lifecycle	Prod. Mgmt	MDG	Sales	All
Compliance	Finance	MDG	Legal	Ops
Manufacturing	Planning	MDG	Eng.	Finance
Pricing	Finance	CFO	Sales	MDG

CRITICAL RULES

- No bulk updates without Guardian approval
- No Studio field creation without documentation
- No field renaming (Studio limitation)
- Metric fields read-only for non-Engineering
- Lifecycle transitions are one-way
- Imperial derives from Metric

Development Rules & Field Naming

Code architecture, naming conventions, and known gotchas

FIELD NAMING CONVENTION

PATTERN	ORIGIN	RULE
`x_studio_*`	Odoo Studio	Read-only in code
`x_kebony_*`	Python (bol_report)	Native codebase field
`kebony_*`	Python (manufacturing)	Newer naming standard
Standard Odoo	Core	Follow Odoo conventions

CODE ARCHITECTURE RULES

- **No duplicate logic** — all metric computations centralised in `KebonyMetricsMixin`
- **Studio fields documented exactly as named** — typos in `x_studio_word_species` and `x_studio_customer_catagory` are permanent
- **No hidden logic** — all derived fields reproducible from base geometry
- **All computed fields must declare dependencies** via `@api.depends`

IMPORT & MIGRATION REFERENCE

SHEET	PURPOSE	RECORDS
MASTERDATA_METRIC_import	Metric products (81 cols)	147
MASTERDATA_IMPERIAL_import	Imperial products	117
import_master item	Master Item grouping	~70
import_master variant	Master Variant grouping	~80

KNOWN FIELD NAME MAPPINGS

ACTUAL TECHNICAL NAME	OLD / SIMPLIFIED NAME
`x_studio_master_item_code`	x_master_item_code
`x_studio_master_variant_code_1`	x_master_variant_code
`x_studio_word_species`	x_wood_species
`x_studio_many2one_field_3oj…`	Application
`x_studio_many2one_field_6up…`	Shape Value
`x_studio_pcslayer`	x_pcs_per_layer

> Studio-generated field names with random suffixes are permanent Odoo identifiers. Always use the exact technical name.

Strategic Impact

Why Product Master Data is infrastructure — not just a catalogue

1. Cubic-Based Margin Logic

Every financial KPI reconciles to m³ through the normalised volume ratio

2. Dryer-Centric Manufacturing

Planning family links products to process parameters and bottleneck constraints

3. FIFO Integrity

Lot-valuated tracking with mandatory lot assignment prevents cost mixing

4. Multi-Entity Operations

Shared product masters across BNV (manufacturing), INC (distribution), Holding

5. Analytical Accounting

Product category and classification drive GL routing and analytic distribution

6. Trade Compliance

HS codes, commodity codes, and country of origin are first-class governed fields

7. No SKU Explosion

Treatment at lot level, no dimensional variants, three-level hierarchy — ~490 products instead of ~11,200

Design principle: Product Master Data is infrastructure. It must be treated with the same rigour as financial chart of accounts or manufacturing Bill of Materials.